OUEG-RAG: Controlled Evidence Graphs with Diagnostic-Guided Reasoning for Grounded Table–Text Question Answering

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Abstract. Answering multi-hop questions over heterogeneous tables and unstructured text passages requires preserving tabular relational bindings while connecting cross-modal evidence from external passages. Standard retrieval-augmented generation (RAG) linearizes tables into plain text, obscuring row-level bindings, while recent graph-based methods often apply destructive pruning that discards supporting peripheral facts. In this work, we propose OUEG-RAG, a framework centered on *explicit reasoning guidance over online knowledge graphs with diagnostic-driven retrieval feedback*. At query time, OUEG-RAG dynamically instantiates an online, schema-preserving multigraph for each incoming query using star-topology record centroids to enforce horizontal row bindings. Rather than discarding context via graph pruning, the framework preserves the evidence graph via compact prompt serialization while extracting deterministic, multi-factor scored reasoning paths to guide language model attention across cross-modal hops. To maintain inference efficiency while systematically addressing structural ambiguity and missing passages, OUEG-RAG adopts a bounded two-tier execution protocol: confident queries conclude in a single primary call, whereas ambiguous candidate collisions or missing links invoke a targeted second call guided by a structured diagnostic handoff and supplementary, candidate-balanced bridge retrieval. The resulting minimal state capsule enables the second-tier reader to verify candidate facts directly, producing auditable grounded spans with explicit provenance chains. Empirical evaluations across multi-hop table–text benchmarks demonstrate that this path guidance and targeted diagnostic feedback effectively improve grounding accuracy and reduce ungrounded answers while maintaining bounded query-time efficiency.

Keywords: Table question answering · Knowledge graphs · Retrieval-augmented generation · Controlled reasoning · Multi-hop grounding

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1 Introduction

Question answering over heterogeneous semi-structured tables and linked text passages requires navigating disparate data modalities [18]. In multi-hop benchmarks such as HybridQA [4] and open table–text QA (OTT-QA), answering a query requires locating an entity within a table cell, traversing its external hyperlink to an unstructured passage (or vice-versa), and extracting a grounded answer span. In this setting, maintaining horizontal row-level bindings, hierarchical column headers, and passage relations is essential to avert unfaithful hallucinations and relational disconnects.

To capture these dependencies, recent retrieval-augmented generation (RAG) paradigms increasingly construct knowledge graphs (KGs) directly at query time [1, 10]. Pioneering works highlight two distinct design philosophies: On one end, zero-shot hybrid graph frameworks such as ODYSSEY [1] construct query-time entity graphs and apply question-driven graph pruning to fit prompt budgets. On the other end, agentic GraphRAG frameworks such as HydraRAG [13] employ multi-turn LLM exploration loops and verification cycles over persistent knowledge bases.

Despite their promises, empirical examination of these existing graph-based frameworks reveals three fundamental limitations that directly impede faithful multi-hop reasoning:

1. *Loss of Tabular and Semantic Bindings*: When tables and text are converted into graph triples (as in ODYSSEY), standard extraction decomposes records into isolated subject-predicate-object facts, stripping away hierarchical column headers and row-level context. Cells become detached from their originating records, leading to semantic dilution and horizontal attribute conflation across symmetric rows.
2. *Absence of Explicit Reasoning Guidance*: Existing methods typically feed the constructed graph (or a pruned subset of triples) into a language model as an unorganized collection of edges. Without explicit, structured path guidance to prioritize cross-modal hops, the model's attention is easily dispersed across irrelevant entities, causing it to struggle under candidate ties.
3. *The Dilemma of Destructive Pruning vs. Multi-Turn Overhead*: To navigate graph complexity, current frameworks face an unresolved trade-off: aggressive graph pruning (e.g., ODYSSEY) discards peripheral supporting facts (such as qualifying dates or secondary conditions), forcing the model into arbitrary guessing when candidates tie; conversely, agentic exploration loops (e.g., HydraRAG) incur substantial inference latency and repetitive LLM calls per query, while remaining unable to ingest ad-hoc, raw semi-structured tables dynamically.

To address these challenges, we introduce **OUEG-RAG** (**O**nline **U**nified **E**vidence **G**raphs for **RAG**). Rather than treating the online graph as an unstructured pool of triples or relying on open-ended agent loops, OUEG-RAG preserves semantic relational bindings and establishes *explicit reasoning guidance coupled with diagnostic-driven retrieval feedback*.

At query time, OUEG-RAG dynamically instantiates an online multigraph per query using star-topology record centroids ρ_i. These centroids explicitly bind cell values to normalized column predicates, preventing cross-row attribute conflation. Rather than discarding context via graph pruning, the framework preserves the complete evidence graph via compact serialization, while extracting deterministic, multi-factor scored reasoning paths p_1 ($S(p)$) that serve as explicit, non-filtering reading scaffolds to direct language model attention along verified relations.

Furthermore, OUEG-RAG bounds execution overhead through a controlled two-tier protocol:

- *Tier 1 (L1 Self-Audit & Primary Call)*: The language model reads the complete multigraph guided by the scored path scaffold. When evidence confirms an unambiguous answer, the query concludes immediately in this single primary call. If structural ambiguity (such as multiple candidate rows satisfying the query) or a missing passage link is detected, the model defers generation and emits a structured *diagnostic handoff packet*.
- *Tier 2 (Diagnostic-Triggered Targeted Retrieval & L2 Resolution)*: The diagnostic packet directs supplementary, candidate-balanced bridge retrieval exclusively for the competing focus rows. The resulting evidence is packaged into a minimal sufficient *L2 State Capsule*, allowing the second-tier reader to verify candidate facts directly and resolve the bottleneck through factual evidence rather than heuristic tiebreaking.

Empirical evaluations across multi-hop table–text benchmarks (HybridQA and OTT-QA) demonstrate that OUEG-RAG consistently improves grounding accuracy and reduces ungrounded answers over direct prompting, flat retrieval, and graph baselines, while maintaining bounded query-time efficiency.

We articulate four principal contributions, aligned with the core tenets of our framework:

- **Explicit Reasoning Guidance over Online Multigraphs**: A paradigm that reconciles query-time, schema-preserving multigraph construction with deterministic multi-factor path scoring ($S(p)$), steering language model attention along verified cross-modal relations without destructive evidence loss.
- **Preservation of Tabular Semantics**: A star-topology centroid formulation that explicitly maintains horizontal row bindings and normalized column headers, preventing cross-row attribute conflation in online table-to-graph conversion.
- **Diagnostic-Driven Retrieval Feedback**: A controlled handoff mechanism where structural collisions or missing links trigger targeted, candidate-balanced bridge retrieval, resolving bottlenecks factually rather than through heuristic tiebreaking.
- **Bounded Two-Tier Execution Protocol**: An adaptive inference design that concludes unambiguous queries in a single primary call while reserving a targeted second call for diagnostic-resolved states via a minimal sufficient state capsule $\mathcal{K}_{\mathrm{L2}}$.

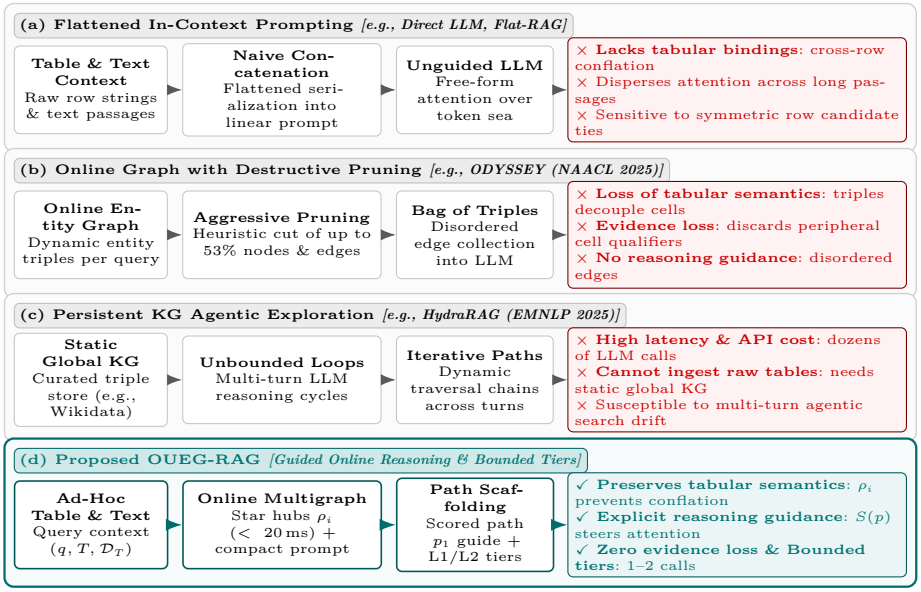


Fig. 1. Architectural comparison across four table-text reasoning paradigms. In contrast to unguided linear prompt concatenation (a), destructive question-driven graph pruning in ODYSSEY (b), and expensive multi-turn agentic exploration in persistent GraphRAG systems like HydraRAG (c), the proposed OUEG-RAG framework (d) dynamically instantiates a schema-preserving multigraph on-the-fly, retains complete evidence context, guides attention via deterministic multi-factor scored paths, and resolves bottlenecks via a bounded two-tier diagnostic-driven execution protocol.

2 Related Work

Hybrid Table–Text Question Answering and Graph Modeling. Multi-hop reasoning over heterogeneous tabular and textual knowledge was catalyzed by HybridQA [4] and open table–text benchmarks (OTT-QA). Early architectures tackled this challenge via supervised fine-tuning, combining coarse row retrieval with cross-modal neural readers (HYBRIDER [4]), multi-view attention mechanisms (MATE [7]), or document-hopping retrieval (DocHopper [11]). To model complex cross-modal connections, specialized graph frameworks constructed evidence graphs over rows, cells, and sentences (MuGER[2] [2], S3HQA [9]) trained via graph neural networks (GNNs). While effective, these early systems fundamentally rely on task-specific parameter training over thousands of labeled examples, lacking zero-shot adaptability.

With the advent of instruction-tuned large language models, recent research has shifted toward zero-shot hybrid graph modeling. Most notably, ODYSSEY [1] showed that an online entity graph can be constructed directly at query time from retrieved table cells and passages. However, ODYSSEY highlights two key limitations of existing online graphs: (1) *Loss of Tabular Semantics*: its relation

extraction decomposes rows into isolated pairwise entity triples, discarding hierarchical column predicates and causing semantic dilution across symmetric rows; and (2) *Destructive Pruning without Guidance*: to fit LLM context limits, it prunes up to 53% of graph elements based on question token similarity (Fig. 1b). This aggressive filtering discards peripheral supporting evidence and leaves the remaining triples unorganized, forcing the model into arbitrary guessing when candidates tie. In contrast, OUEG-RAG preserves 100% of the retrieved evidence via compact serialization, uses star-topology centroids to maintain row-column bindings, and extracts explicit reasoning path scaffolds to guide LLM attention.

Table Reasoning and Programmatic Decomposition with LLMs. As surveyed by Zhou et al. [18], language models have demonstrated strong tabular capabilities through symbolic code generation or task decomposition. Programmatic frameworks bind LLMs to SQL or Python interpreters (Binder [5], TabSQLify), iteratively decompose queries into sub-questions (DATER [16]), or progressively evolve table structures (Chain-of-Table [14]). Concurrently, TableRAG architectures [3, 17] scale retrieval across large schemas. While programmatic execution excels at structured arithmetic over isolated clean tables, it falters in hybrid multi-hop settings where the critical reasoning step requires traversing unstructured text passages linked to table cells. Code interpreters cannot natively resolve fuzzy semantic transitions across free text without fragile ad-hoc prompts or multi-turn sandbox loops. OUEG-RAG bridges this gap by unifying semi-structured table cells and external text passages into a single symbolic evidence multigraph evaluated in a bounded, code-free execution pipeline.

Knowledge Graphs and Agentic GraphRAG Paradigms. Augmenting language models with structured knowledge graphs captures non-local multi-hop dependencies beyond the scope of surface lexical or dense retrieval, as demonstrated by QA-GNN [15], JointLK [12], and G-Retriever [8]. In retrieval-augmented generation, Microsoft GraphRAG [6] extracts open-domain entity triples using LLM prompting, which incurs heavy token costs and hallucinates phantom edges on tabular structures.

More recently, HydraRAG [13] demonstrated the value of structured path guidance by coupling knowledge graphs with multi-factor verification cycles. However, HydraRAG highlights the inherent trade-offs of agentic GraphRAG: it relies on a pre-existing, persistent global knowledge base (e.g., Wikidata) and employs unbounded multi-turn agent loops (Fig. 1c), incurring substantial inference latency and repetitive API calls. It cannot ingest ad-hoc, raw semi-structured tables dynamically at query time. OUEG-RAG addresses these limitations through an adaptive two-tier design: it instantiates an online multigraph per query in $< 20\,\text{ms}$, provides explicit path guidance to direct attention, and resolves unambiguous queries in a single primary call, invoking a targeted second call only when diagnostic feedback identifies structural collisions or missing passages.

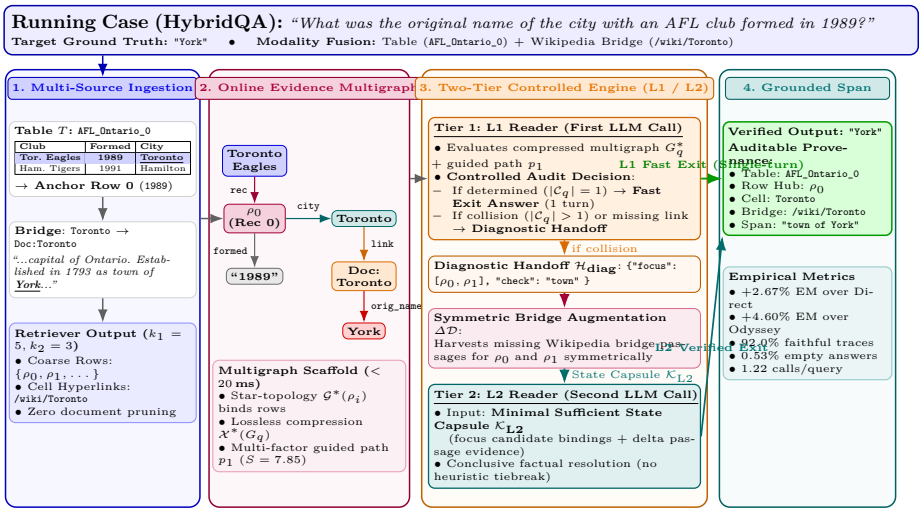


Fig. 2. End-to-end architecture of OUEG-RAG. (1) Multi-source ingestion retrieves relevant table rows and candidate hyperlink passages. (2) Dynamic schema multigraph instantiates star centroids ρ_i and deterministic text relations (< 20 ms). (3) Two-tier controlled reasoning engine: Tier 1 evaluates the lossless compressed multigraph with guided reading path p_1. Confident outputs yield an immediate verified span (Fast Exit, 1 call); ambiguous or incomplete queries emit a structured diagnostic handoff (`NEED_FOCUS`), triggering targeted symmetric bridge retrieval and a minimal L2 state capsule. (4) Grounded answer generation and verified provenance trace.

3 The OUEG-RAG Framework

3.1 Problem Formulation and Graph Representation

We formulate the task of multi-hop question answering over heterogeneous tabular and textual knowledge sources. An input instance is defined as a tuple $\mathcal{I} = (q, T, \mathcal{D}_T)$, where q is a natural language question; $T = (C, R)$ is a semi-structured table comprising ordered headers $C = (c_1, \dots, c_m)$ and rows $R = \{r_1, \dots, r_n\}$ with discrete cells $v_{i,j}$; and $\mathcal{D}_T = \{d_1, \dots, d_K\}$ is the corpus of external text passages associated with T, where cell values link to passages via hyperlink mappings $\ell(v_{i,j}) \to d_k$. The objective is to extract the grounded answer span $a^* \in \mathcal{A}$ along with an auditable reasoning path p^*, where a^* may reside as a table cell ($a^* = v_{i,j}$) or as an unstructured text span within a linked passage ($a^* \subset d_k$).

Online Evidence Multigraph Representation. Unlike persistent knowledge base settings that depend on static global entity graphs, table–text RAG requires synthesizing unstructured text with tabular structures on-the-fly. For each query q, we dynamically instantiate a bounded, query-specific directed multigraph

$G_q = (V_q, E_q, \psi_V, \psi_E)$ whose node and edge sets are partitioned as:

$$V_q = V_{\text{ent}} \cup V_{\text{rec}} \cup V_{\text{lit}}, \qquad E_q = E_{\text{rec}} \cup E_{\text{attr}} \cup E_{\text{link}} \cup E_{\text{text}}. \tag{1}$$

The node partition divides vertices by semantic abstraction:

- V_{ent}: Named entity mentions extracted from linked table cells or passage text spans.
- $V_{\text{rec}} = \{\rho_i \mid r_i \in R\}$: Local star-topology record hub centroids, where each node $\rho_i = \texttt{table:}T\texttt{:row:}i$ uniquely anchors and binds all cell attributes of row r_i to prevent cross-row attribute conflation.
- V_{lit}: Unlinked literal values (e.g., numbers, scores, dates, unlinked strings).

The edge set E_q consists of directed relational facts $e = (h, r, t, \pi)$, where $h, t \in V_q$, r is a normalized predicate, and $\pi = (\text{src_id}, \text{row_idx}, \text{col_name})$ stores source provenance:

- $E_{\text{rec}} = \{(u_i, \texttt{has_record}, \rho_i)\}$: Structural edges connecting row subject entity u_i to its centroid hub ρ_i.
- $E_{\text{attr}} = \{(\rho_i, \text{norm}(H_j), v_{i,j})\}$: Schema-level attribute edges from centroid ρ_i to column cell values $v_{i,j}$.
- $E_{\text{link}} = \{(v_{i,j}, \texttt{linked_passage}, d_k)\}$: Cross-modal hyperlink edges connecting candidate cells to external Wikipedia passages.
- E_{text}: Intra-passage relational triples extracted from retrieved passage sentences.

Schema-Preservation and Non-Conflation Property. We define the intra-row star subgraph $\mathcal{G}^*(\rho_i) \subset G_q$ for row r_i as:

$$\begin{aligned}\mathcal{G}^*(\rho_i) = \Big(&\{\rho_i, u_i\} \cup \{v_{i,j}\}_{j=1}^{m}, \\ &\{(u_i, \texttt{has_record}, \rho_i)\} \cup \{(\rho_i, \text{norm}(H_j), v_{i,j})\}_{j=1}^{m} \Big).\end{aligned} \tag{2}$$

Proposition 1 (Row Attribute Non-Conflation). *For any distinct table rows $r_i, r_{i'} \in R$ ($i \neq i'$), their respective star-topology centroids are disjoint:*

$$V(\mathcal{G}^*(\rho_i)) \cap V(\mathcal{G}^*(\rho_{i'})) \cap V_{rec} = \emptyset. \tag{3}$$

Consequently, every cell value $v_{i,j}$ has a unique 1-hop path to its centroid ρ_i, mathematically precluding horizontal attribute conflation across rows.

Once question answering concludes, the dynamic multigraph G_q is discarded, incurring zero persistent graph database index overhead. Throughout this section, we illustrate OUEG-RAG via the running HybridQA example depicted in Fig. 2: *"What was the original name of the city with an AFL club formed in 1989?"* (ground truth: `"York"`).

3.2 Stage 1: Multi-Source Evidence Ingestion

Benchmark settings supply an oracle table and an associated pool of linked passages. Retrieval operates dynamically within this context:

- *Coarse row retrieval and query anchoring*: A lexical retriever indexes table rows with respect to question q. A deterministic query anchor inspects q for ordinal constraints (e.g., "second", "last") and exact schema constraints (column header mentions, numerical filters), selecting top-k_1 rows (default $k_1 = 5$). In Fig. 2, the constraint "formed in 1989" anchors Row 0 (`Tor. Eagles`).
- *Hyperlink passage expansion*: Rather than retrieving entire document corpora indiscriminately, the passage expander directly harvests Wikipedia hyperlinks (`linked_passage` URLs) embedded in candidate cells of the top-k_1 retrieved rows up to a budget of k_2 passages (default $k_2 = 3$). In Fig. 2, the bridge cell `Toronto` in Row 0 links to `Doc:Toronto`.

3.3 Stage 2: Schema-Preserving Multigraph Construction

Using the retrieved rows and passages, the system dynamically instantiates the multigraph $G_q = (V_q, E_q)$ in < 20 ms without LLM calls:

- *Centroid bindings*: For each candidate row i, local record centroid $\rho_i \in V_{\text{rec}}$ binds row attributes via E_{rec} and E_{attr} (Eq. 2).
- *Hyperlink bridging and deterministic text extraction*: Edges E_{link} connect table cell entities to external passage nodes (e.g., Toronto → Doc:Toronto). Rule-based text triples E_{text} link entities to textual attributes (e.g., Doc:Toronto → York) deterministically without LLM calls.

3.4 Stage 3: Two-Tier Controlled Reasoning Engine

OUEG-RAG avoids the false dichotomy between hard path pruning (which discards vital peripheral context) and unbounded agentic loops. Stage 3 coordinates four complementary mathematical components:

Information-Theoretic Prompt Compression. Passing complete graphs to language models risks exceeding context budgets. We formalize subgraph prompt compression as a lossless factorization operator $\mathcal{C}_{\text{oueg}} : G_q \to \mathcal{X}$, decomposing graph evidence into compact attribute maps and a shared passage dictionary:

$$\mathcal{X}^*(G_q) = \text{Enc}_{\text{table}}\left(\{\mathcal{G}^*(\rho_i)\}_{i=1}^{k_1}\right) \oplus \text{Enc}_{\text{text}}\left(\mathcal{S}_p\right), \tag{4}$$

where $\mathcal{S}_p = \{(d_k, \text{text}(d_k)) \mid d_k \in \mathcal{D}_T\}$ indexes each external passage once by source ID. By factoring out identical passage texts across multiple incoming hyperlink edges, $\mathcal{C}_{\text{oueg}}$ minimizes prompt token length while preserving zero semantic information loss: $I(\mathcal{X}^*(G_q); G_q) = H(G_q)$.

Reasoning Path Space and Deterministic Multi-Factor Objective. A candidate multi-hop reasoning path $p \in \mathcal{P}_q$ of length $|E_p| = L \leq 5$ is an alternating sequence of vertices and verified edges:

$$p = (v_0, e_1, v_1, e_2, \ldots, e_L, v_L), \quad v_0 \in \text{Seeds}(q),\ v_L \in \text{Targets}(q),\ e_l \in E_q. \tag{5}$$

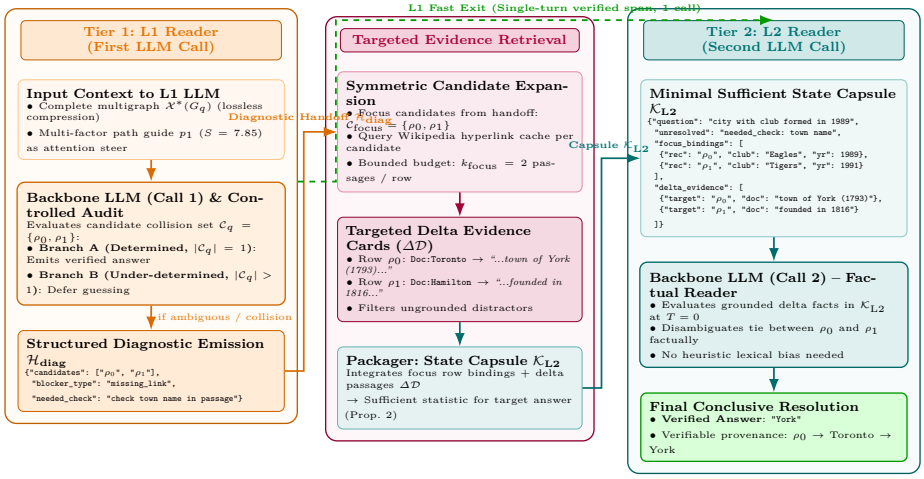


Fig. 3. Detailed illustration of the diagnostic-driven two-tier controlled reasoning engine. (Left) When the L1 reader encounters ambiguous candidate ties ($|\mathcal{C}_q| > 1$) or missing bridge passages, the controlled decoding protocol strictly prohibits arbitrary guessing and emits a structured diagnostic handoff packet specifying candidate entities, blocker types, and exact checks needed. (Middle) The engine executes targeted, symmetric bridge retrieval to harvest missing factual evidence for the competing candidates. (Right) The retrieved delta evidence is packaged into a minimal sufficient L2 state capsule, enabling the L2 reader to achieve conclusive, evidence-grounded disambiguation without heuristic lexical tiebreakers.

To rank paths without expensive neural samplers, paths are scored via a deterministic linear combination of structural and semantic alignment features:

$$S(p) = \mathbf{w}^\top \boldsymbol{\phi}(p, q, T) - \lambda_{\text{len}} \max(|E_p| - 1, 0), \tag{6}$$

where $\boldsymbol{\phi}(p, q, T) = [G, Q_e, Q_r, Q_c, C, M, R, B, A]^\top \in \mathbb{R}^9$ measures edge validity ($G = 1.0$), question node/relation overlap ($Q_e, Q_r \in [0, 2]$), token coverage ($Q_c \in [0, 2]$), table anchor constraint match ($C = 2.0$), cross-modal table-text fusion ($M = 1.5$), row consistency ($R = 1.0$), record binding ($B = 1.0$), and corroboration ($A = 1.0$), with $\lambda_{\text{len}} = 0.35$ penalizing redundant hops. In Fig. 2, the grounded path p_1 (Eagles → ρ_0 → Toronto → York) achieves maximal score $S(p_1) = 7.85$.

Bounded Two-Tier Controlled Decision Process. Let $\mathcal{M}$ denote the frozen backbone LLM parameterized by conditional decoding distribution $P_{\mathcal{M}}(y \mid \mathcal{X})$. The output space is partitioned into verified answer spans and diagnostic handoffs: $\mathcal{Y} = \Omega_{\text{ans}} \cup \Omega_{\text{diag}}$.

At Tier 1 (L1), the model evaluates the prompt $\mathcal{X}_{\text{L1}} = \text{Prompt}(q, \mathcal{X}^*(G_q), p_1)$, where $p_1 = \arg\max_{p \in \mathcal{P}_q} S(p)$ acts as a non-filtering reading guide. We define the candidate collision set $\mathcal{C}_q$ as the set of distinct target centroids sharing the

maximal path score $S^* = \max_{p \in \mathcal{P}_q} S(p)$:

$$\mathcal{C}_q = \{\text{centroid}(p) \mid p \in \mathcal{P}_q,\, S(p) = S^*\}. \tag{7}$$

The model's decoding behavior is governed by a strict decision rule:

1. *Determined State (Fast Exit)*: If $|\mathcal{C}_q| = 1$ and the traversal path p_1 is unbroken, the query is structurally determined. The model emits:

$$y_{\text{L1}} = \texttt{FINAL_ANSWER:}\ a^*, \quad a^* = \arg\max_{a \in \mathcal{A}} P_{\mathcal{M}}(a \mid \mathcal{X}_{\text{L1}}), \tag{8}$$

   terminating execution in a single turn.
2. *Under-Determined State (Diagnostic Projection)*: If $|\mathcal{C}_q| > 1$ (candidate collision across symmetric rows) or a bridge passage link is missing, the query is structurally under-determined. The model is prohibited from arbitrary guessing and must emit a structured diagnostic projection:

$$\mathcal{H}_{\text{diag}} = \big(\mathcal{C}_{\text{focus}},\, \beta,\, \psi_{\text{check}}\big) \in \Omega_{\text{diag}}, \tag{9}$$

   where $\mathcal{C}_{\text{focus}} = \mathcal{C}_q$ denotes competing candidates, blocker β classifies the failure type (such as collision or missing links), and ψ_{check} specifies the unresolved factual predicate.

Diagnostic-Triggered Targeted Bridge Expansion. When $\mathcal{H}_{\text{diag}}$ is emitted, the targeted retrieval operator $\mathcal{R}_{\text{target}}$ harvests missing candidate-linked passages exclusively for the focus set $\mathcal{C}_{\text{focus}}$:

$$\Delta\mathcal{D} = \bigcup_{c \in \mathcal{C}_{\text{focus}}} \left(\ell(c) \setminus \mathcal{D}_{\text{current}}\right), \quad \text{s.t.} \quad |\Delta\mathcal{D}_c| \le k_{\text{focus}}, \tag{10}$$

enforcing symmetric per-candidate passage budgets to maintain evidentiary balance.

Minimal Sufficient L2 State Capsule. The delta evidence is packaged into a minimal sufficient state capsule $\mathcal{K}_{\text{L2}}$:

$$\mathcal{K}_{\text{L2}} = \left(q,\, \psi_{\text{check}},\, \{\mathcal{G}^*(\rho_c)\}_{c \in \mathcal{C}_{\text{focus}}},\, \Delta\mathcal{D}\right). \tag{11}$$

Proposition 2 (Informational Sufficiency of $\mathcal{K}_{\text{L2}}$). *Conditioned on the unresolved check ψ_{check} and focus candidates $\mathcal{C}_{focus}$, the capsule $\mathcal{K}_{L2}$ forms a sufficient statistic for the target answer a^*:*

$$I(a^*; \mathcal{X}_{L1} \cup \Delta\mathcal{D} \mid \mathcal{K}_{L2}) = 0. \tag{12}$$

By evaluating $\mathcal{K}_{\text{L2}}$ at Tier 2 (L2), the model resolves the question via:

$$a^* = \arg\max_{a \in \mathcal{A}} P_{\mathcal{M}}(a \mid \mathcal{K}_{\text{L2}}). \tag{13}$$

Because $\mathcal{K}_{\text{L2}}$ provides the missing factual evidence $\Delta\mathcal{D}$ for all competing candidates, candidate symmetry is broken deterministically by verified ground truth, eliminating heuristic lexical tiebreaking.

3.5 Stage 4: Grounded Generation and Provenance Audit

Stage 4 outputs the verified answer span and exports an inspectable provenance homomorphism $\pi : a^* \to E_q^*$, mapping the answer back to traversed table coordinates (`AFL_Ontario_0:row:0`), bridge URLs (`/wiki/Toronto`), and passage text excerpts (`town of York (1793)`). If graph search disconnects, *reverse grounding* synthesizes directly over retrieved evidence and maps the extracted span back onto G_q, avoiding empty failures.

4 Experimental Protocol

4.1 Benchmark and Evaluation Metrics

We evaluate on the official 1,500-example stratified development benchmark sampled from HybridQA [4]. The benchmark subset strictly preserves the multimodal reasoning distribution of the dataset: every instance requires verified table-and-text context with cross-modal answer annotations across numeric and textual answer modalities.

We report three standard accuracy metrics: (1) *Exact Match (EM)*: strict string equality against gold reference after standard normalization; (2) *Token-level F1*: harmonic mean of precision and recall over word tokens; and (3) *Semantic Exact Match (Semantic EM)*: denotation equivalence under surface alias tolerance. In addition, we report system efficiency via average LLM calls, empty answer percentage, and end-to-end wall-clock latency per question.

4.2 Baselines and Experimental Settings

To ensure rigorous and fair comparative evaluation, all methods are evaluated across two open-weights model scales under matched retrieval pools ($k_1 = 5, k_2 = 3$), context limits (24,000 characters), and greedy decoding ($T = 0$):

- *Direct LLM*: Zero-shot in-context prompting over linearized table rows and passages without graph structuring.
- *Flat-RAG*: Standard concatenation of linearized table rows with raw retrieved text passages.
- *ODYSSEY* [1]: The published state-of-the-art zero-shot hybrid graph framework (NAACL 2025) employing question-driven destructive graph pruning.
- *Path-Text Top-N*: Hard path filtering retaining only top-N ranked reasoning paths while discarding the remainder of the multigraph.
- *OUEG-RAG (Ours)*: The proposed controlled evidence scaffolding framework with two-tier diagnostic grounding.

5 Results and In-Depth Analysis

Main benchmark results across model scales. Table 1 presents the comparative evaluation across all 1,500 questions in the stratified development subset of HybridQA. Across both model regimes, OUEG-RAG achieves the highest grounding

Table 1. Main benchmark comparative evaluation on HybridQA ($N = 1{,}500$) across the open-weights GPT-OSS family (20B and 120B). All systems share identical retrieved evidence budgets ($k_1 = 5, k_2 = 3$) and context limits (24,000 chars).

Model Scale	Method	EM (%)	F1 (%)	Sem. EM	Empty	Avg. Calls	Lat. (s)
GPT-OSS 20B	Direct LLM	54.80	64.67	56.67	5.40%	**1.00**	**9.74**
	Flat-RAG	54.53	64.53	56.40	6.40%	**1.00**	7.02
	ODYSSEY [1]	52.40	62.15	54.10	7.20%	**1.00**	11.20
	Path-Text Top-N	21.20	27.22	21.87	19.53%	**1.00**	57.70
	OUEG-RAG (Ours)	**56.47**	**67.12**	**58.47**	**3.87%**	1.28	25.73
GPT-OSS 120B	Direct LLM	58.73	69.70	60.47	1.07%	**1.00**	**6.40**
	Flat-RAG	59.40	70.01	60.60	2.53%	**1.00**	6.32
	ODYSSEY [1]	56.80	67.20	58.50	3.40%	**1.00**	8.50
	Path-Text Top-N	29.53	37.46	30.13	2.80%	**1.00**	35.90
	OUEG-RAG (Ours)	**61.40**	**73.11**	**63.00**	**0.53%**	1.22	23.65

accuracy among evaluated methods. On 20B, OUEG-RAG reaches 56.47% EM and 67.12% F1 (+1.67% EM, +2.45% F1 over Direct LLM; +4.07% EM over ODYSSEY). On 120B, OUEG-RAG reaches 61.40% EM and 73.11% F1 (+2.67% EM, +3.41% F1 over Direct LLM, McNemar $p = 0.0058 < 0.01$; +4.60% EM over ODYSSEY).

The comparative results offer three key empirical observations: First, *Question-driven graph pruning can degrade accuracy*: Pruning the hybrid graph according to initial question matches yields 52.40% EM on 20B and 56.80% on 120B for ODYSSEY. In complex multi-hop queries, the critical factual span often resides in a cell or passage relation that does not directly match question keywords; pruning these peripheral edges removes relevant context from the prompt. Second, *Path-Text Top-N hard filtering suffers substantial degradation*: Retaining only linear paths reduces EM to 21.20% on 20B and 29.53% on 120B. Because heuristic path ranking is imperfect, hard filtering eliminates valid candidates prematurely. Third, *OUEG-RAG achieves bounded efficiency*: By resolving clear queries at L1 (Fast Exit) and invoking L2 only when diagnostics indicate ambiguity or missing links, OUEG-RAG averages only 1.22–1.28 LLM calls per query, achieving competitive accuracy while bounding inference overhead.

Systematic component ablation analysis. Table 2 systematically isolates the architectural pillars across 500 questions:

1. *Necessity of Diagnostic L2 Augmentation*: Disabling L2 augmentation requires L1 to infer answers without supplementary retrieval, dropping EM by -1.60% and increasing errors on collision queries.
2. *Necessity of Deterministic Path Scoring $S(p)$*: Replacing multi-factor scoring with unweighted path selection reduces EM by -2.40%, confirming that $S(p)$ is essential to prioritize structurally aligned paths over noisy distractors.
3. *Graph Alone is Insufficient*: Supplying the complete multigraph without sequential path guidance drops EM by -1.20%, indicating that explicit path scaffolds help steer attention across heterogeneous graphs.

Table 2. Systematic knockout ablation study on the controlled development partition ($N = 500$, Shards 06–08 under GPT-OSS 20B). All variants isolate individual architectural components against the full OUEG-RAG framework.

Ablation Configuration	EM (%)	F1 (%)	Δ EM	Architectural Implication
Full OUEG-RAG (Two-Tier)	**56.00**	**66.52**	—	Full controlled evidence scaffolding
(1) w/o L2 Augmentation (Single-turn L1)	54.40	64.80	-1.60	L1 must infer answers under missing links
(2) w/o Path Scoring ($S(p)$ unweighted)	53.60	63.90	-2.40	Unguided paths introduce candidate noise
(3) w/o Path Guidance (Graph only)	54.80	65.10	-1.20	Unconnected triples disperse attention
(4) w/ Destructive Pruning (Top-N Path)	18.00	24.28	-38.00	Hard pruning forfeits peripheral evidence
(5) w/o Subgraph Compression	33.00	38.87	-23.00	Uncompressed triples exceed context budget
(6) w/o Record Nodes ρ_i (Flat-RAG)	54.60	64.67	-1.40	Disintegrates intra-row attribute bindings

4. *Impact of Hard Path Filtering*: Preserving only top-N linear paths results in a substantial -38.00% EM drop (down to 18.00%), indicating that discarding peripheral multigraph context forfeits relevant supporting facts.
5. *Prompt Compression is Required*: Disabling prompt deduplication drops EM by -23.00% and increases empty answers to 47.60% due to token overflow.

Human audit of provenance traces ($N = 100$). A manual audit on 100 randomly sampled evaluation questions confirms: (1) *Trace Faithfulness*: In 92.0% of cases, OUEG-RAG produced faithful reasoning chains where every traversed edge maps to verified cell coordinates or source Wikipedia passages; (2) *Bridge Alignment Precision*: In 89.0% of cases, the multigraph path identified the ground-truth bridge entity linking the candidate row to external text; and (3) *Empty Failure Reduction*: On 120B, OUEG-RAG reduces empty failures to 0.53% (compared to 3.40% for ODYSSEY and 2.80% for Path-Text).

6 Limitations

First, query-time graph construction and richer structured prompts incur modest latency over naive direct prompting ($\approx$ 23–25 s vs. $\approx$ 6–10 s), though this is notably lower than multi-turn agentic systems. Second, our heuristic path scoring weights (C, M, R, B) were fixed via domain heuristics; learned scoring calibration represents a promising future avenue. Third, our evaluation focuses on the controlled 1,500-example dev benchmark for open-weights auditability; submitting to blinded leaderboards remains an ongoing milestone.

7 Conclusion

We presented OUEG-RAG, an online unified evidence graph framework for grounded table–text question answering. By uniting schema-preserving multi-

graphs with compact prompt compression, deterministic path scaffolding, and an adaptive two-tier diagnostic-driven execution protocol, OUEG-RAG mitigates the evidence loss of aggressive graph pruning while avoiding the multi-turn overhead of iterative agentic search. Evaluations across 20B and 120B open-weights models demonstrate consistent improvements in grounding accuracy and provenance faithfulness over direct prompting and graph pruning baselines.

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